- **L3 MIASHS/Ingémath/METIS**
- **Université Paris Cité**
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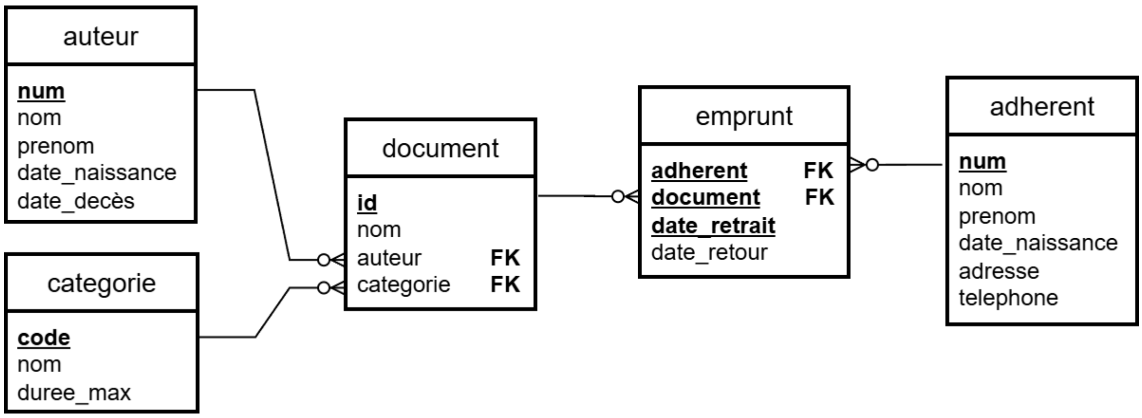
# Examen du 9 janvier 2026 (3H)

## Exercise 1

Here is the relational model of a media library's information system.




This IS records document loans by media library members. Documents fall into several categories (`livre`, `BD`, `CD`, `DVD`). Each category has a specific maximum loan term expressed in days.

- The attributes `auteur.num`, `categorie.code`, `document.id` and `adherent.num` are integers as well as the attributes that reference them.
- The attributes `emprunt.date_retrait` and `emprunt.date_retour` are of type `DATE` and store the dates on which a member borrowed and returned a document. **When a document has not yet been returned**, the attribute `emprunt.date_retour` is empty.
- The attributes `auteur.date_naissance`, `auteur.date_deces` and `adherent.date_naissance` are of type `DATE`.
- The attribute `categorie.duree_max` is an integer that stores the maximum number of days a document in the category can be borrowed for.

### Questions

1. Write the SQL query to define the `emprunt` table, taking into account the relational model and the previous instructions.
2. Write a query in SLQ that lists the names of documents authored by `Jules Verne`.
3. Write a query in SLQ that lists the number of documents in the media library for each author. Each line should include the author's number, the author's name, and the number of documents.
4. Write an SLQ query that lists all the information about members who have borrowed at least one document that has not yet been returned. Each member should only appear once.
5. Write an SQL query that lists the `id` of documents in the category named `BD` that have been borrowed at least 10 times.
6. Write an SQL query that lists the numbers of members who have never borrowed the document with the `id` 200.
7. Write an SQL query that returns the average number of loans per document for each category during the year 2025. Each row should only contain the category code and the average number.

**Note** : the expression `emprunt.date_retrait BETWEEN '2025-01-01' AND '2025-12-31'` allows you to test whether a loan was taken out during the year 2025.

8. Write a query in SQL, without subqueries, without CTE (without `WITH`) and without INTERSECT, that lists the numbers of members who have borrowed at least once the document with `id 100` **and** at least once the document with `id 200`. A member number must appear at most once.
9. Write an SQL query that indicates the document borrowed by the largest number of different members.
10. (**BONUS**) Explain why the attribute `emprunt.date_retrait` is part of the identifier for the relationship `emprunt`.

## Exercise 2

We want to design a database to manage tennis court reservations by members of a sports club.

The following must be recorded in the database :

- Club members with a unique number, last name, and first name.
- Club coaches with a unique number, last name, and first name.
- Facilities with a unique number, name, and type of facility.
- Facility types (indoor court, outdoor court, trainer, etc.) with a unique code, name, and description.

The future reservation entry interface will look like this :

- Enter the member number.
- Enter the date, time, and duration of the desired reservation.
- Enter the desired type of facility. After entering this field, the system will display the facilities available for the desired type in the requested time slot, and **the member will select a facility**. If no facility is selected or available, the reservation will not be created.
- Enter the request for a trainer : “yes/no.” If the member answers “yes,” the system will display the list of available trainers with their hourly rates, and **the member will choose a trainer**. If no trainer is chosen or available, the reservation will not be created unless “no” is answered to the request for a trainer.

The **hourly rate for a trainer** depends on the trainer and the type of facility where they work.

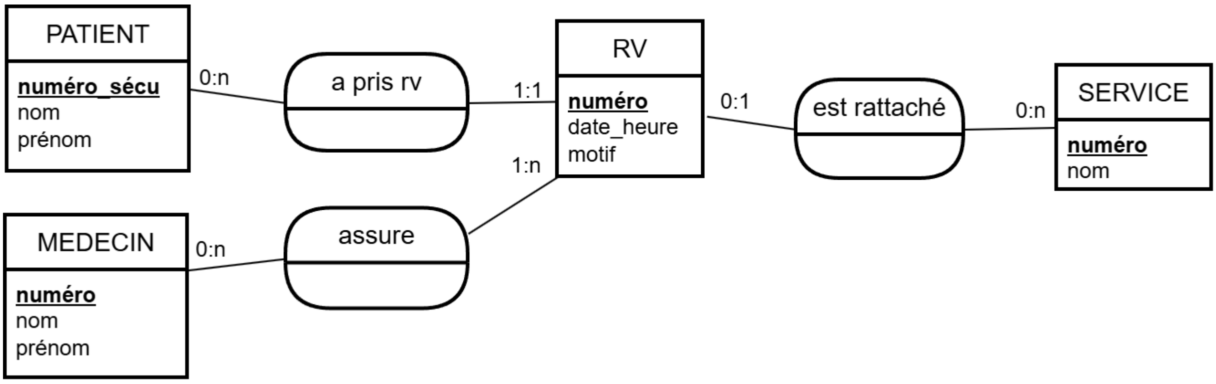
### Questions

1. Write an E/A diagram for this IS, modeling reservations using an entity. Indicate the cardinalities and justify those that add a constraint to the model.
2. Write an advanced E/A diagram for this IS, modeling reservations without using an entity. Indicate the cardinalities (no justification required). Verify that a member can reserve the same facility multiple times. Add constraints to the diagram if necessary.

## Exercise 3

We want to create a database to record appointments for a medical clinic.

The design team has developed the following Conceptual Data Model :




Additional information :

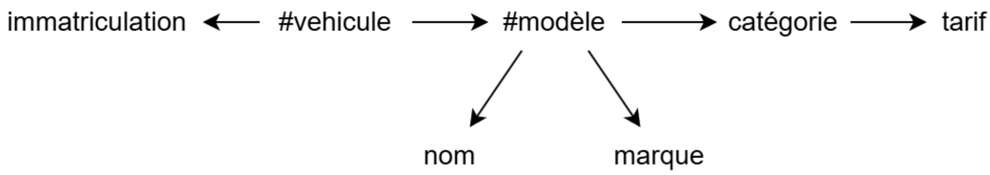
- In certain specific cases, an appointment requires the presence of several doctors.
- Some appointments are linked to a department, others are not.

### Questions

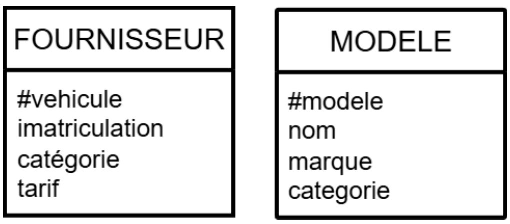
1. Convert this E/A diagram into a relational model. No justification required.
2. Which foreign keys will require a `NOT NULL` or `UNIQUE` constraint when defining the tables in SQL ? Justify your answer.

## Exercise 4

The study of an IS led to this diagram of the DFs :




The design team has provided you with this relational diagram :

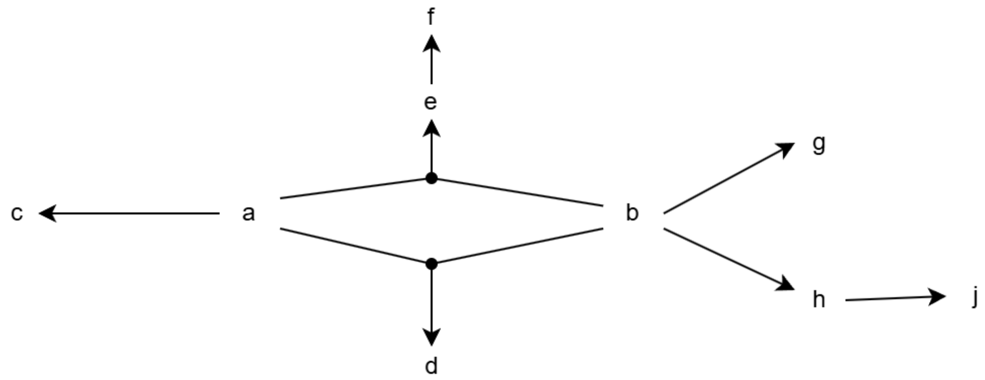



### Questions

1. Determine whether this relational schema causes DF losses.
2. For each relationship in this schema, determine the keys and check whether the relationship is in FNBC ; if not, give the optimal normal form that it verifies.
3. For tables that are not FNBC, illustrate redundancies with examples of tuples.

## Exercise 5

Here is the diagram of the DFs of an IS :




### Questions

1. Propose a FN3 diagram for this IS.
2. Show that this diagram is in FN3.
3. Is this diagram in FNBC ? Justify your answer.